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Automated showerhead tilt adjustment

Abstract

In some examples, an automated tilting system is provided for adjusting an orientation of a component in a substrate processing chamber. The automated tilting system comprises at least one tilt adjustment motor arranged to cooperate with the component and be connected to a portion of the component by a coupling. The coupling is configured such that automated rotational motion by the at least one tilt adjustment motor imparts corresponding axial movement, relative to the at least one tilt adjustment motor or a datum structure, to the connected portion of the component to adjust the orientation of the component in the processing chamber.

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Background/Summary

CLAIM OF PRIORITY (1) This application is a U.S. National Stage Filing under 35 U.S.C. 371 from International Application No. PCT/US2021/036808, filed on Jun. 10, 2021, and published as WO 2021/262446 A1 on Dec. 30, 2021, which claims the benefit of priority to U.S. Provisional Patent Application Ser. No. 63/042,980, filed on Jun. 23, 2020, each of which is incorporated by reference herein in its entirety.

FIELD

(1) The present disclosure relates generally to systems, apparatus, and methods for automated tilt adjustment of a showerhead component in substrate processing systems.

BACKGROUND

(2) A conventional showerhead and cooling plate assembly in a substrate processing system typically includes a series of threaded rods or screws that can be adjusted manually to tilt the showerhead cooling plate into a desired orientation. A particular orientation may be desired after planned maintenance or during a showerhead replacement, for example. The orientation operation typically involves a manual gap check of a substrate-support assembly each time an initial adjustment in orientation is made, followed by some form of metrology to verify a “fine tuning” of the tilting operation until the desired orientation is met. This manual adjustment usually requires an appropriately qualified expert to be physically present on site and typically involves several hours of cooling down and heating up of the showerhead components. Tool operators cannot change tilt orientations quickly, yet imperfect or rushed tilting can lead to undesired wafer scrap. Moreover, cooling plate components are often of brazed form and the extra complexity and inspection of these components can add manufacturing time and cost.

(3) The background description provided herein is to generally present the context of the disclosure. It should be noted that the information described in this section is presented to provide the skilled artisan some context for the following disclosed subject matter and should not be considered as admitted prior art. More specifically, work of the presently named inventors, to the extent it is described in this background section, as well as aspects of the description that may not

otherwise qualify as prior art at the time of filing, are neither expressly nor impliedly admitted as prior art against the present disclosure.

SUMMARY

(4) The present disclosure relates generally to systems, apparatus, and methods for automated tilt adjustment of a showerhead component in substrate processing systems.

(5) In some examples, an automated tilting system for adjusting an orientation of a component in a substrate processing chamber is provided. An example automated tilting system comprises at least one tilt adjustment motor arranged to cooperate with the component; the at least one tilt adjustment motor connected directly or indirectly to a connected portion of the component by a coupling, the coupling configured such that automated rotational motion by the at least one tilt adjustment motor imparts corresponding axial movement, relative to the at least one tilt adjustment motor or a datum structure, to the connected portion of the component thereby to adjust the orientation of the component in the processing chamber.

(6) In some examples, the component is a showerhead component or a cooling plate component. In some examples, the coupling includes an axial plunge coupler-screw interface. In some examples, the axial plunge coupler-screw interface decouples axial load from an associated tilt adjustment motor. In some examples, the axial plunge coupler-screw interface includes a ball-end screw received in a spherical cup. In some examples, the automated tilting system further comprises a bellows interposed between the component and the processing chamber. In some examples, the bellows at least partially provides a vacuum seal between the component and the processing chamber. In some examples, the automated tilting system further comprises one or more load compensators. In some examples, the component is, or includes, a friction stir welded component.

(7) In some examples, a substrate processing system comprises an automated tilting system for adjusting an orientation of a component in a substrate processing chamber, the automated tilting system comprising: at least one tilt adjustment motor arranged to cooperate with the component; the at least one tilt adjustment motor connected directly or indirectly to a connected portion of the component by a coupling, the coupling configured such that automated rotational motion by the at least one tilt adjustment motor imparts corresponding axial movement, relative to the at least one tilt adjustment motor or a datum structure, to the connected portion of the component thereby to adjust the orientation of the component in the processing chamber. In some examples, the automated tilt adjustment system of the substrate processing system includes any one or more of the elements summarized above.

(8) In some examples, a non-transitory computer-readable storage medium is provided. An example computer-readable storage medium includes instructions which, when executed by a controller, cause the controller to communicate with at least one tilt adjustment motor arranged to cooperate with a component in a substrate processing chamber, wherein the at least one tilt adjustment motor is connected directly or indirectly to a connected portion of the component by a coupling, the coupling configured such that automated rotational motion of the at least one tilt adjustment motor imparts corresponding axial movement, relative to the at least one tilt adjustment motor or a datum structure, to the connected portion of the component: and selectively engage the at least one tilt adjustment motor to adjust an orientation of the component.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) Some embodiments are illustrated by way of example and not limitation in the views of the accompanying drawings:

(2) FIG. 1 is a diagram showing an example arrangement of a processing chamber within which some examples of the present disclosure may be employed, according to some example

embodiments.

(3) FIG. 2 is a representative diagram illustrating a friction stir welding arrangement in accordance with one embodiment.

(4) FIG. 3 is a representative diagram illustrating an automated tilting system in accordance with an example embodiment.

(5) FIG. 4 is a block diagram illustrating an example of a controller upon which one or more example embodiments may be implemented, or by which one or more example embodiments may be controlled.

(6) FIG. 5 is a flow chart illustrating a method in accordance with one embodiment.

DETAILED DESCRIPTION

(7) The description that follows includes systems, arrangements, methods, techniques, and computing machine program products that embody illustrative embodiments of the present disclosure. In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of example embodiments. It will be evident, however, to one skilled in the art that the present disclosure may be practiced without these specific details.

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(9) With reference now to FIG. 1, an example arrangement **100** of a plasma-based processing chamber is shown. The present subject matter may be used in a variety of semiconductor manufacturing and wafer processing operations, but in the illustrated example, the plasma-based processing chamber is described in the context of plasma-enhanced or radical-enhanced Chemical Vapor Deposition (CVD) or Atomic Layer Deposition (ALD) operations. The skilled artisan will also recognize that other types of ALD processing techniques are known (e.g., thermal-based ALD operations) and may incorporate a non-plasma-based processing chamber. An ALD tool is a specialized type of CVD processing system in which ALD reactions occur between two or more chemical species. The two or more chemical species are referred to as precursor gases and are used to form a thin film deposition of a material on a substrate, such as a silicon wafer as used in the semiconductor industry. The precursor gases are sequentially introduced into an ALD processing chamber and react with a surface of the substrate to form a deposition layer. Generally, the substrate repeatedly interacts with the precursors to deposit an increasingly thick layer of one or more material films on the substrate. In certain applications, multiple precursor gases may be used to form various types of film or films during a substrate manufacturing process. The gases may be extremely chemically or corrosively aggressive.

(10) FIG. 1 is shown to include a plasma-based processing chamber **102** in which a showerhead **104** and a substrate-support assembly **108** or pedestal are disposed. The showerhead **104** may include a component such as a. showerhead electrode, or a chandelier showerhead, for example. The showerhead **104** may include a component such as cooling plate **120** described further below. Other showerhead components are possible. Some examples include systems, apparatus, and methods for automated tilt adjustment of a showerhead component in substrate processing systems, and more particularly to automated fine tuning of such components.

(11) Typically, the substrate-support assembly **108** seeks to provide a substantially-isothermal surface and may serve as both a heating element and a heat sink for a substrate **106**. The substrate-support assembly **108** may comprise an Electrostatic Chuck (ESC) in which heating elements are included to aid in processing the substrate **106**, as described above. The substrate **106** may include a wafer comprising, for example, elemental-semiconductor materials (e.g., silicon (Si) or

germanium (Ge)) or compound-semiconductor materials (e.g., silicon germanium (SiGe) or gallium arsenide (GaAs)). Additionally, other substrates include, for example, dielectric materials such as quartz, sapphire, semi-crystalline polymers, or other non-metallic and non-semiconductor materials.

(12) In operation, the substrate **106** is loaded through a loading port **110** onto the substrate-support assembly **108**. A gas line **114** can supply one or more process gases (e.g., precursor gases) to the showerhead **104**. In turn, the showerhead **104** delivers the one or more process gases into the processing chamber **102**. A gas source **112** (e.g., one or more precursor gas ampules) to supply the one or more process gases is coupled to the gas line **114**. In some examples, an RF (radio frequency) power source **116** is coupled to the showerhead **104**. In other examples, a power source is coupled to the substrate-support assembly **108** or ESC.

(13) Prior to entry into the showerhead **104** and downstream of the gas line **114**, a point-of-use (POU) and manifold combination (not shown) controls entry of the one or more process gases into the processing chamber **102**. In the case of a processing chamber **102** used to deposit thin films in a plasma-enhanced ALD operation, precursor gases may be mixed in the showerhead **104**.

(14) In operation, the processing chamber **102** is evacuated by a vacuum pump **118**. RF power is capacitively coupled between the showerhead **104** and a lower electrode (not shown explicitly) contained within or on the substrate-support assembly **108**. RF power may be inductively or transformer coupled in some examples. The substrate-support assembly **108** is typically supplied with two or more RF frequencies. For example, in various embodiments, the RF frequencies may be selected from at least one frequency at about 1 MHz, 2 MHz, 13.56 MHz, 27 MHz, 60 MHz, and other frequencies as desired. A coil designed to block or partially block a particular RF frequency can be designed as needed. Therefore, particular frequencies discussed herein are provided merely for ease in understanding. The RF power is used to energize the one or more process gases into a plasma in the space between the substrate **106** and the showerhead **104**. The plasma can assist in depositing various layers (not shown) on the substrate **106**. In other applications, the plasma can be used to etch device features into the various layers on the substrate **106**. RF power is coupled through at least the substrate-support assembly **108**. The substrate-support assembly **108** may have heaters (not shown in FIG. 1) incorporated therein. The detailed design of the processing chamber **102** may vary.

(15) In some examples, friction stir welding (FSW) is used to manufacture or join pedestal or showerhead parts such as a faceplate, a cooling plate, and a pedestal stem. The manufacture or joining of other components is possible. The methods and articles of manufacture described herein may exhibit improved thermal and mechanical properties as compared to articles made using conventional techniques such as vacuum brazing and electron beam welding. Friction stir welded components may also be more manufacturable in terms of time, cost, and reduced complexity especially for sophisticated substrate components for substrate processing tools.

(16) FSW addresses several limitations of conventional techniques. Such limitations may include weld cracking and porosity resulting in a reduced heat transfer within pedestal or showerhead components and a diminished efficiency of a processing chamber to which such a pedestal or showerhead has been fitted.

(17) With reference to FIG. 2, an arrangement **200** for friction stir welding is provided. A first component **202** of a showerhead **104** or cooling plate **120** may be joined to a second component **204** in the manner shown. The illustrated first component **202** may include an edge portion of a cooling plate **120** (see FIG. 3 below), disposed for example in the processing chamber **102**. The illustrated second component **204** may include an edge portion of another component of the showerhead **104**, for example a platen disposed in the same zone as the first component **202**. As shown, a downward force is exerted on the first component **202** and second component **204** to be joined together by a rotating FSW tool **206**. The FSW tool **206** is advanced in a welding direction **208** between the edge portions of the first component **202** and the second component **204**. The

FSW tool **206** includes a shoulder **210** and a pin **212** and creates an FSW region **214** as the FSW tool **206** advances through the material of the first component **202** and the second component **204** to be joined. The FSW region **214** so formed includes a nugget **216** that joins the first component **202** and the second component **204** together. The nugget **216** has an advancing side **218** and a retreating side **220** corresponding to the rotational direction **222** of the FSW tool **206**.

(18) FIG. **3** is a part-sectional diagram illustrating an automated tilting system **300** for adjusting an orientation of a showerhead component, such as a showerhead **104** or cooling plate **120** in a processing chamber **102**, for example. The orientation may include a planar orientation. In the illustrated example, the cooling plate **120** is connected to a showerhead **104** but other arrangements are possible. In some examples, the cooling plate **120** is a friction stir welded cooling plate **120** or includes friction stir welded components. The cooling plate **120** may include components joined together, or be manufactured at least in part, by an FSW tool **206** in an FSW method described above with reference to FIG. **2**. The cooling plate **120** may include one or more channels **306** or conduits for coolant or gas.

(19) As described more fully below, the automated tilting system **300** acts in an automated manner to adjust a planar orientation (or tilt) of the cooling plate **120**, and thus the showerhead **104** connected to it, with respect to the processing chamber **102** and the substrate **106** supported within the processing chamber **102**. To this end, the automated tilting system **300** includes an arrangement of tilt adjustment motors **302**. In the illustrated example, three tilt adjustment motors **302** are provided. Only one motor **302** is fully visible in the view of FIG. **3**, a second motor **302** is shown in sectional view, while a third motor is not visible by dint of the line of section. Other numbers and/or configurations of tilt adjustment motors **302** are possible. An automated tilting system **300** comprising three tilt adjustment motors **302** is considered to provide an acceptable degree of tilt fine-tuning capability.

(20) Each of the tilt adjustment motors **302** includes a mounting plate **303** connected to a motor base **308**. Each motor base **308** is secured to the cooling plate **120** by one or more fasteners **310**. Each of the tilt adjustment motors **302** has a motor shaft **314**. Each motor shaft **314** is connected to and held within a coupling **318**. The coupling may comprise several parts described in more detail below. In some examples, the motor shaft **314** is splined and held captive in a complementary recess in an upper section **320** of the coupling **318**. The splines transmit torque applied from the tilt adjustment motors **302** to the upper section **320** of the coupling **318**. Torque is transmitted to a lower section **322** of the coupling **318** by internal keys or connectors **324**.

(21) The lower section **322** of the coupling **318** interacts with a screw **326** to provide what is termed an axial plunge coupler-screw interface. In the illustrated example, an upper end **328** of the screw **326** is of box section such that the upper end **328** can receive torque from the lower section **322** of the coupling **318**. Further embodiments may include hex or other sections, internal or external coupling arrangements, splines, and tripod joints. The lower section **322** of the coupling **318** includes a complementary box-section recess **330** recess. The box-section recess **330** is deeper than the length of the upper end **328** of the screw **326** and this provides a free “plunge space”. The screw **326** can thus receive torque from the tilt adjustment motors **302**, yet move axially (i.e., plunge) within the box-section recess **330** in the lower section **322** of the coupling **318**. Axially, therefore, the tilt adjustment motors **302** are free of the cooling plate **120** and the showerhead **104** connected to it. The tilt adjustment motors **302** therefore carry no axial load.

(22) An operatively lower end of the screw **326** includes a ball **332**. The ball **332** cooperates with a socket clamp **334**. The ball-and-socket arrangement imparts a degree of freedom to the automated tilting system **300** such that the cooling plate **120** and showerhead **104** assembly can assume planar orientations that are not necessarily horizontal or orientations that are “tilted” or unaligned with respect to the processing chamber **102** or other datum structure. Other coupling arrangements to this end are possible, for example spherical or other kinds of bearing types or assemblies, such as needle point or jewel bearings. A datum structure may include an external base or supporting

component, for example. Other datum structures are possible.

(23) The socket clamp **334** is fixed to the processing chamber **102** and the screw **326** is threaded at thread **338** in the showerhead **104**. Thus, selective rotational movement of the screw **326** by the tilt adjustment motors **302** causes the screw **326** to move up or down in an axial or vertical direction relative to the relevant tilt adjustment motors **302**, and the showerhead **104**, and draw the connected portion of the processing chamber **102** nearer or further away from the showerhead **104**. The term “vertical” herein is not necessarily intended to be limited to a strict “geometrically vertical” value (although that value is included within its ambit), but is intended to include examples that are generally vertical or upright.

(24) A planar orientation of the showerhead **104** and cooling plate **120** can be adjusted by controlled action of the tilt adjustment motors **302** working together. Precise degrees of control imparted by the rotary encoder **304** enable the automated tilting system **300** to adjust the orientation of the showerhead **104** and cooling plate **120** in a “fine-tuned” manner. In an example arrangement comprising three tilt adjustment motors **302**, the automated tilting system **300** provides automated three-axis tilting capability. Other arrangements that include fewer or more tilt adjustment motors **302** are possible.

(25) The processing chamber **102** includes a bellows **312**. The bellows **312** define an extendible and compressible vacuum seal or zone between the processing chamber **102** and the showerhead **104** and can accommodate movement between the processing chamber **102** and the showerhead **104**. The bellows **312** allow the showerhead **102** to move without breaking a vacuum seal. In some examples, other seal means may be provided, for example an arrangement including one or more O-rings.

(26) In some examples, the processing chamber **102** is attached to an external base or supporting component by fasteners **316**. In these examples, the processing chamber **10** may be said to be fixed (datum structure) in which case an adjusted showerhead **104** or cooling plate **120** would move relative to the processing chamber **102**. In other examples, the opposite situation occurs. In the illustrated example, the weight of the showerhead **104** and a loading from a differential pressure (i.e. outside atmosphere versus an inside “vacuum” load) is supported by the three ball-and-socket clamps (i.e. the ball **332** and the socket clamp **334**), but is at least partially offset by one or more load compensation springs **336**. Other compensation mechanism or arrangements (for example gas springs) are possible. The degree of load compensation may be selected in view of the strength of any vacuum suction inside the bellows **312**.

(27) The rotary encoder **304** may include or be controlled by one or more controller **400**. In some examples, a controller **400** provides intelligent or “smart” control capability. Intelligent control may include for example tilting between substrate processing steps, more concise and user-friendly tilt adjustment. Some examples include preemptive adjustment of tilt to mitigate the effects of wear of components on process results. Some examples enable tracking of how much tilt would be used over time for any tool, set of tools, design, and so forth. Some examples store previous showerhead tilt settings so when the next showerhead is installed, any offsets that fingerprint the tool can be applied to “estimate” the correct tilt and increase likelihood of first pass success.

(28) FIG. 4 is a block diagram illustrating an example controller **400** by which one or more systems or methods described herein may be implemented or controlled. In alternative embodiments, the controller **400** may operate as a standalone device or may be connected (e.g., networked) to other machines. In a networked deployment, the controller **400** may operate in the capacity of a server machine, a client machine, or both in server-client network environments. In an example, the controller **400** may act as a peer machine in a peer-to-peer (P2P) (or other distributed) network environment. Further, while only a single controller **400** is illustrated, the term “machine” (controller) shall also be taken to include any collection of machines (controllers) that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein, such as via cloud computing, software as a service (SaaS), or

other computer cluster configurations. In some examples, and referring to FIG. 4, a non-transitory machine-readable medium includes instructions **424** that, when read by a controller **400**, cause the controller to control operations in methods comprising at least the non-limiting example operations described herein.

(29) Examples, as described herein, may include, or may operate by logic, a number of components, or mechanisms. Circuitry is a collection of circuits implemented in tangible entities that include hardware (e.g., simple circuits, gates, logic, etc.). Circuitry membership may be flexible over time and underlying hardware variability. Circuitries include members that may, alone or in combination, perform specified operations when operating. In an example, hardware of the circuitry may be immutably designed to carry out a specific operation (e.g., hardwired). In an example, the hardware of the circuitry may include variably connected physical components (e.g., execution units, transistors, simple circuits, etc.) including a Computer-Readable Medium physically modified (e.g., magnetically, electrically, by moveable placement of invariant massed particles, etc.) to encode instructions of the specific operation. In connecting the physical components, the underlying electrical properties of a hardware constituent are changed (for example, from an insulator to a conductor or vice versa) The instructions enable embedded hardware (e.g., the execution units or a loading mechanism) to create members of the circuitry in hardware via the variable connections to carry out portions of the specific operation when in operation. Accordingly, the Computer-Readable Medium is communicatively coupled to the other components of the circuitry when the device is operating. In an example, any of the physical components may be used in more than one member of more than one circuitry. For example, under operation, execution units may be used in a first circuit of a first circuitry at one point in time and reused by a second circuit in the first circuitry, or by a third circuit in a second circuitry, at a different time.

(30) The machine (e.g., computer system) controller **400** may include a hardware Processor **402** (e.g., a central processing unit (CPU), a hardware processor core, or any combination thereof), a GPU **432** (graphics processing unit), a main memory **404**, and a static memory **406**, some or all of which may communicate with each other via an interlink **408** (e.g., a bus) The controller **400** may further include a display device **410**, an alphanumeric input device **412** (e.g., a keyboard), and a UI navigation device **414** (e.g., a mouse or other user interface). In an example, the display device **410**, alphanumeric input device **412**, and UI navigation device **414** may be a touch screen display. The controller **400** may additionally include a mass storage device **416** (e.g., drive unit), a signal generation device **418** (e.g., a speaker), a network interface device **420**, and one or more sensors **430**, such as a Global Positioning System (GPS) sensor, compass, accelerometer, or another sensor. The controller **400** may include an output controller **428**, such as a serial (e.g., universal serial bus (USB)), parallel, or other wired or wireless (e.g., infrared (IR), near field communication (NFC), etc.) connection to communicate with or control one or More peripheral devices (e.g., a printer, card reader, etc.).

(31) The mass storage device **416** may include a machine-readable medium **422** on which is stored one or more sets of data structures or instructions **424** (e.g., software) embodying or utilized by any one or more of the techniques or functions described herein. The instructions **424** may as shown also reside, completely or at least partially, within the main memory **404**, within the static memory **406**, within the hardware processor **402**, or within the GPU **432** during execution thereof by the controller **400**. In an example, one or any combination of the hardware processor **402**, the GPU **432**, the main memory **404**, the static memory **406**, or the mass storage device **416** may constitute the machine-readable medium **422**.

(32) While the machine-readable medium **422** is illustrated as a single medium, the term “machine-readable medium” may include a single medium, or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) configured to store the one or more instructions **824**.

(33) The term “machine-readable medium” may include any medium that can store, encode, or carry instructions **424** for execution by the controller **400** and that cause the controller **400** to perform any one or snore of the techniques of the present disclosure, or that can store, encode, or carry data structures used by or associated with such instructions **424**. Non-limiting machine-readable medium examples may include solid-state memories, and optical and magnetic media. In an example, a massed machine-readable medium comprises a machine-readable medium **422** with a plurality of particles having invariant (e.g., rest) mass. Accordingly, massed machine-readable media are not transitory propagating signals. Specific examples of massed machine-readable media may include non-volatile memory, such as semiconductor memory devices (e.g., electrically programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM)) and flash memory devices; magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The instructions **424** may further be transmitted or received over a communications network **426** using a transmission medium via the network interface device **420**.

(34) Some examples herein include methods. With reference to FIG. 5, a method **500** of adjusting an orientation of a component in a substrate processing chamber is provided. The method **500** comprises, at operation **502**, arranging at least one tilt adjustment motor to cooperate with the component; at operation **504**, connecting each tilt adjustment motor directly or indirectly to a connected portion of the component by a coupling, the coupling configured such that automated rotational motion by the at least one tilt adjustment motor imparts corresponding axial movement, relative to the at least one tilt adjustment motor or a datum structure, to the connected portion of the component thereby to adjust the orientation of the component in the processing chamber; and, at operation **506**, selectively engaging the at least one tilt adjustment motor to adjust the orientation of the component.

(35) Although examples have been described with reference to specific example embodiments or methods, it will be evident that various modifications and changes may be made to these embodiments without departing from the broader scope of the embodiments. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. The accompanying drawings that form a part hereof, show by way of illustration, and not of limitation, specific embodiments in which the subject matter may be practiced. The embodiments illustrated are described in sufficient detail to enable those skilled in the art to practice the teachings disclosed herein. Other embodiments may be utilized and derived therefrom, such that structural and logical substitutions and changes may be made without departing from the scope of this disclosure. This detailed description, therefore, is not to be taken in a limiting sense, and the scope of various embodiments is defined only by the appended claims, along with the full range of equivalents to which such claims are entitled.

(36) Such embodiments of the inventive subject matter may be referred to herein, individually and/or collectively, by the term “invention” merely for convenience and without intending to voluntarily limit the scope of this application to any single invention or inventive concept if more than one is in fact disclosed. Thus, although specific embodiments have been illustrated and described herein, it should be appreciated that any arrangement calculated to achieve the same purpose may be substituted for the specific embodiments shown. This disclosure is intended to cover any and all adaptations or variations of various embodiments. Combinations of the above embodiments, and other embodiments not specifically described herein, will be apparent to those of skill in the art upon reviewing the above description.

Claims

1. An automated tilting system for adjusting an orientation of a component in a substrate processing chamber, the automated tilting system comprising: at least one tilt adjustment motor arranged to

cooperate with the component; the at least one tilt adjustment motor connected directly or indirectly to a connected portion of the component by a coupling, the coupling configured such that automated rotational motion by the at least one tilt adjustment motor imparts corresponding axial movement, relative to the at least one tilt adjustment motor or a datum structure, to the connected portion of the component thereby to adjust the orientation of the component in the processing chamber, wherein the coupling comprises: a plunge interface that allows axial movement of a driven member relative to a drive member while transmitting rotational motion between the drive member and the driven member, wherein the driven member is configured to move axially independent of the at least one tilt adjustment motor to decouple axial loads from the at least one tilt adjustment motor, and wherein the component is a showerhead component or a cooling plate component.

2. The automated tilting system of claim 1, wherein the coupling includes an axial plunge coupler-screw interface.

3. The automated tilting system of claim 2 wherein the axial plunge coupler-screw interface decouples axial load from an associated tilt adjustment motor.

4. The automated tilting system of claim 2, wherein the axial plunge coupler-screw interface includes a ball-end screw received in a spherical cup.

5. The automated tilting system of claim 1, further comprising a bellows interposed between the component and the processing chamber.

6. The automated tilting system of claim 5, wherein the bellows at least partially provides a vacuum seal between the component and the processing chamber.

7. The automated tilting system of claim 1, further comprising one or more load compensators.

8. The automated tilting system of claim 1, wherein the component includes a friction stir welded component.
